Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Compute the derivative of the following expressions.

1. $\ln(x)$.
2. $\ln(\sqrt{2x+3})$.
3. $\sqrt{\sin(x)+x^3}$.
4. $\cos(e^{-x})$.

Exercise 1.

Consider the function

$$f(x) = \frac{3x^2 - 2x}{x+1}.$$

1. Compute the derivative of $f(x)$ using the quotient rule.
2. Factor $f(x) = x\left(\frac{3x-2}{x+1}\right)$ and compute the derivative of $f(x)$ using the product rule and the quotient rule.
3. Rewrite $f(x) = (3x^2 - 2x)(x+1)^{-1}$ and compute the derivative of $f(x)$ using the product rule and the chain rule.
4. Explain the relation between these three derivatives.

Exercise 2.

Find the equation $L(x) = mx + b$ of the line tangent to $g(x) = e^{\sin(x)}$ at the point $(\pi, 1)$.